Lecture 1

Linear Transformations & Matrices

Core Definitions

We begin by establishing the properties of fields mapping vector components.

Definition 1.1 : Linear Map

A function $T : V \to W$ between two vector spaces over the same field F is called a **linear transformation** if it satisfies:

1. Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. Homogeneity: $T(cv) = cT(v)$ for all $c \in F$ and $v \in V$

Definition 1.2 : Vector Space

…

Fundamental Theorems

Theorem 1.3 : Rank-Nullity Theorem

Let V and W be vector spaces, where V is finite-dimensional. If $T : V \to W$ is a linear map, then:

$$\dim(\text{null } T) + \dim(\text{range } T) = \dim V$$

Homework Practice

Exercise 1.4 : Identity Dimension Mapping

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find a structured matrix representation M_T and verify the output dimension explicitly:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 1.5 : Identity Dimension Mapping

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find the dimension of the null space.

Solution:

By inspecting the matrix row parameters:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rank is clearly 2 because the two rows are linearly independent. By Rank-Nullity, $\dim(\text{null } T) = 3 - 2 = 1$.

Note 1.6

Remember that every basis is linearly independent.

Warning 1.7

Do **not** confuse image and codomain.

Example 1.8

Let

$$T(x, y) = (x + y, y).$$

Proof.

We proceed by induction...

□

Remark 1.9

The converse is false.

History 1.10

The Rank-Nullity theorem appeared in the nineteenth century.

Lemma 1.11

lemma statement

Proposition 1.12

proposition statement

Claim 1.13

claim statement

Corollary 1.14

corollary statement

Lecture 2

Eigenvalues & Spectral Mapping

Core Definitions

In this session, we analyze how linear transformations scale specific vector subspaces.

Definition 2.1 : Eigenvalues and Eigenvectors

Let $T : V \to V$ be a linear operator on a vector space V over a field F. A non-zero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in F$ such that:

$$T(v) = \lambda v$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector v.

Fundamental Theorems

Theorem 2.2 : Linear Independence of Eigenvectors

Let $T : V \to V$ be a linear operator. If $v_1, v_2, \ldots, v_m$ are eigenvectors corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$, then the set $\{v_1, v_2, \ldots, v_m\}$ is linearly independent.

Homework Practice

Exercise 2.3 : Characteristic Polynomial Evaluation

Consider the 2×2 matrix A representing a linear transformation on $\mathbb{R}^2$:

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

1. Compute the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
2. Find the distinct eigenvalues λ_1 and λ_2 by solving:

$$\lambda^2 - 7\lambda + 10 = 0$$

Lecture 30

Olympiad Inequalities

Core Definitions

Theorem 30.1

There are infinitely many prime numbers.